Aqueous uptake and sublethal toxicity of p,p'-DDE in non-feeding larval stages of Antarctic krill (Euphausia superba).

This study evaluated the toxicological sensitivity of non-feeding larval stages of a key Antarctic species (Antarctic krill, Euphausia superba) to p,p'-dichlorodiphenyl dichloroethylene (p,p'-DDE) exposure. The aqueous uptake clearance rate of 84 mL g(-1) preserved weight (p.w.) h(-1) determined for p,p'-DDE in Antarctic krill larvae is comparable to previous findings for small cold water crustaceans and five times slower than the rates reported for an amphipod inhabiting warmer waters. Natural variations in larval physiology appear to influence contaminant uptake and larval krill behavioural responses, strongly highlighting the importance of time of measurement for ecotoxicological testing. Sublethal narcosis (immobility) was observed in larval Antarctic krill from p,p'-DDE body residues of 0.2 mmol/kg p.w., which is in agreement with findings for adult krill and temperate aquatic species. The finding of comparable body residue-based toxicity of p,p'-DDE between polar and temperate species supports the tissue residue approach for environmental risk assessment of polar ecosystems.